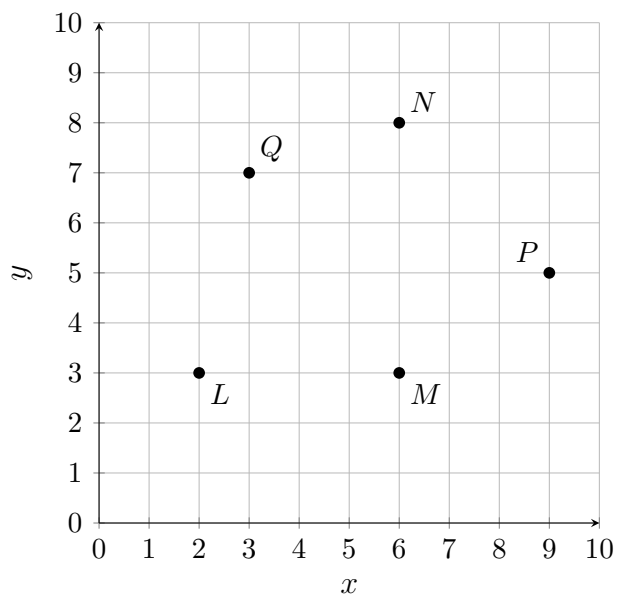


Reading and Plotting Points on a Grid

Naming points, moving on the grid, plotting points, and turning patterns into points

Directions

- An ordered pair is written (x, y) : count *across* first, then *up*. The order is the whole idea.
- Problems 1 to 4 all read from Park Map below. Later problems have their own blank grid to draw on.



Park Map — used by problems 1 to 4. Each labelled dot is a landmark.

1. Use the Park Map. Write the ordered pair for landmark L .

$L = (\rule{1cm}{0.4pt} , \rule{1cm}{0.4pt})$

2. Use the Park Map. Which labelled landmark is farthest to the right?

Answer: _____

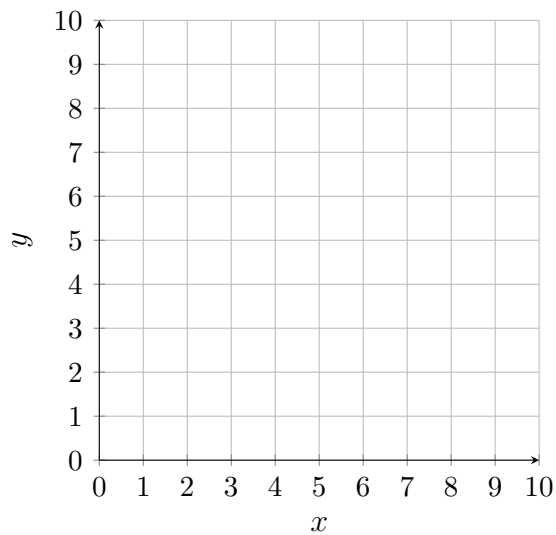
3. Use the Park Map. Start at landmark L , move 4 units to the right, then 5 units up. Write the ordered pair where you land.

(_____ , _____)

4. Use the Park Map. Landmarks L and M sit on the same horizontal line. How far apart are they?

Answer: _____ units

5. Plot and label the points $A(1, 4)$, $B(5, 4)$ and $C(5, 9)$ on the grid below.



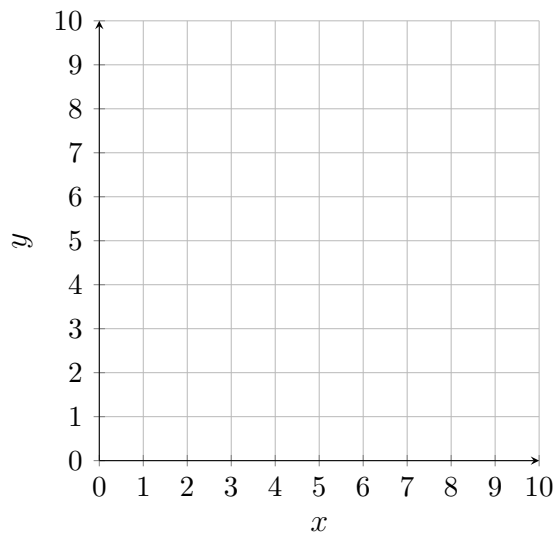
6. A candle is 20 centimetres tall and burns down 3 centimetres each hour. Each hour is recorded as an ordered pair (hour, height in centimetres). Write the ordered pair for hour 5. The hour has been filled in for you.

(5 , _____)

7. Plot $D(2, 2)$, $E(9, 2)$, $F(9, 6)$ and $G(2, 6)$ on the grid, then join them in order to sketch the rectangle.

(a) How many units long is side DE ?

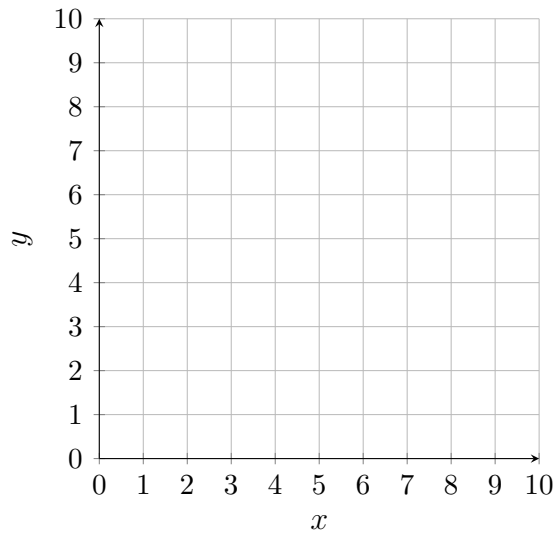
Answer: _____ units



8. A rule says: start at $(0, 1)$, and for every step move 1 unit right and 2 units up.

(a) Write the ordered pair after 4 steps: (_____ , _____)

(b) Plot the first three points of this rule on the grid.



9. The candle from problem 6 gives these points for hours 0 to 4: $(0, 20)$, $(1, 17)$, $(2, 14)$, $(3, 11)$, $(4, 8)$.

(a) Write the mean of the five y -coordinates.

Answer: _____ cm

(b) Write the range of the five y -coordinates.

Answer: _____ cm

10. Two rules build points. Rule 1: the y -coordinate is double the x -coordinate. Rule 2: the y -coordinate is 6 more than the x -coordinate.

(a) Rule 1 at $x = 4$: (4 , _____)

(b) Rule 2 at $x = 4$: (4 , _____)

(c) At which x do the two rules give the same point?

Answer: _____

(d) Explain how you found that x .
